

Gravitational Force

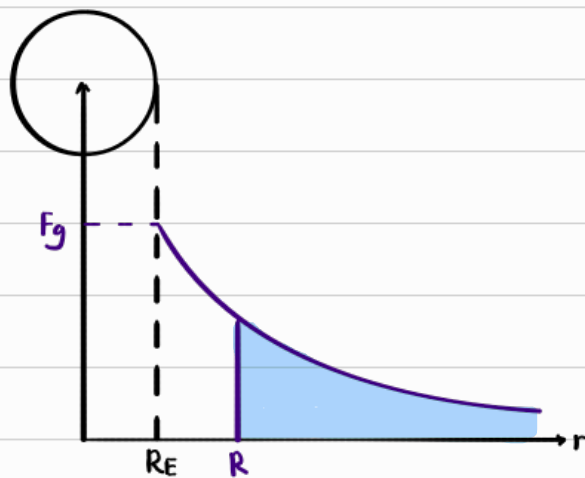
Law of gravitation $\begin{cases} F_g \propto m_1 m_2 \\ F_g \propto \frac{1}{r^2} \end{cases}$

$$\therefore F_g = G \left[\frac{m_1 m_2}{r^2} \right]$$

$$F_g = \frac{GM_1 m_2}{r^2}$$

Gravitational PE

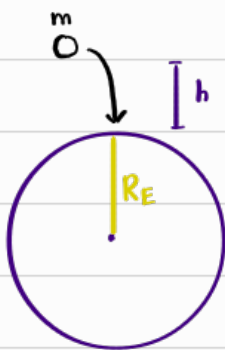
- Work done to bring a mass from infinity to that point



Work done = Area under graph

$$\begin{aligned} u &= \int_{\infty}^R F_g \, dr \\ &= \int_{\infty}^R \frac{GMm}{r^2} \, dr \\ &= GMm \left[-\frac{1}{r} \right]_{\infty}^R \\ &= GMm \left[-\frac{1}{R} + \frac{1}{\infty} \right] \end{aligned}$$

$$u = -\frac{GMm}{R}$$



$$\begin{aligned} \Delta u &= u_f - u_i \\ &= \left(-\frac{GMm}{R_E} \right) - \left(-\frac{GMm}{R_E + h} \right) \\ &= GMm \left[\frac{-(R_E + h) + R_E}{R_E(R_E + h)} \right] \end{aligned}$$

Assume : $h \ll R_E \implies R_E(R_E + h) \approx R_E^2$

$$\begin{aligned} \therefore \Delta u &= -GMm \left[\frac{h}{R_E^2} \right] \\ &= -m \left[\frac{GM}{R_E^2} \right] h \end{aligned}$$

$$\Delta u = -mgh$$

Because m lost GPE
[Became more -ve]

Escape velocity

KE $\rightarrow$ PE

$$\begin{aligned} \frac{1}{2} m v_e^2 &= [u_f - u_i] \\ &= \left[-\frac{GMm}{\infty} - \left(-\frac{GMm}{R_E} \right) \right] \end{aligned}$$

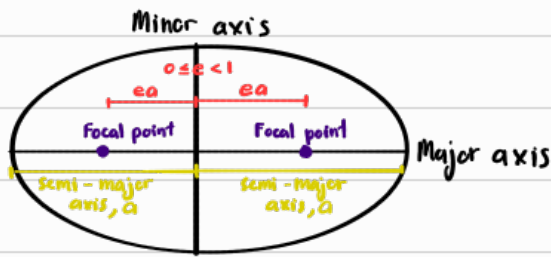
Assume m is on the surface of the earth

$$v_e = \sqrt{\frac{2GM}{R_E}}$$

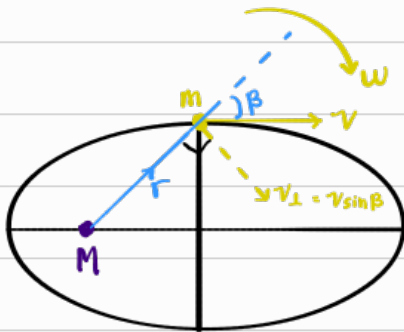
$\approx 11 \text{ km/s}$

Angular Momentum

Orbit is an ellipse



- 1 of the Focal point = position of M [Sun]
- When $e = 0$, ellipse becomes a circle



Angular momentum, $\vec{L} = \vec{r} \times \vec{p}$
 $= \vec{r} \times (m\vec{v})$

$$\vec{L} = m(\vec{r} \times \vec{v})$$

Magnitude of $\vec{L}$, $|\vec{L}| = m[r v \sin \beta]$ (angle btwn $\vec{r}$ & $\vec{v}$)
 $= m r (v_{\perp})$
 $= m r (r \omega)$

$$|\vec{L}| = m r^2 \omega$$

$$|\vec{L}| = I \omega$$

If no external forces, angular momentum will be conserved

$$\sum L_i = \sum L_f$$

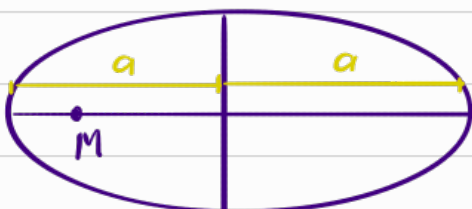
$$m(r v \sin \beta)_i = m(r v \sin \beta)_f$$

$$m(r^2 \omega)_i = m(r^2 \omega)_f$$

Period of orbit

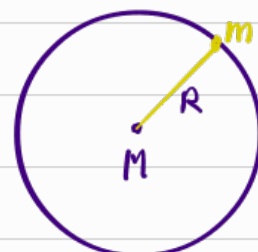
Eclipse

From Kepler's 3rd Law, $T^2 \propto a^3$
 $T^2 = \left(\frac{4\pi^2}{GM}\right) a^3$
 [hours]



Circle

From Kepler's 3rd Law, $T^2 \propto R^3$
 $T^2 = \left(\frac{4\pi^2}{GM}\right) R^3$
 [hours]



Geosynchronous orbit

$T_{\text{satellite orbit}} = T_{\text{M rotation}}$

$$\frac{T_s^2}{r_s^3} = \frac{T_m^2}{R_s^3} \quad [\text{Kepler's 3rd Law}]$$

